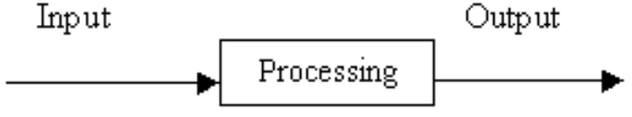
Introduction

System

- A collection of components that work together to realize some objective.
- Basically there are three major components in every system, namely input, processing and output.



1. Input: Involves capturing and assembling elements that enter the system to be processed.
2. Processing: Involves transformation processes that convert input into output. For example mathematical operations.
3. Output: involves transferring elements that have been produced by a transformation process to their ultimate destination.

System Components

i. Resources

Resources can be hardware, software or liveware. Hardware resources may include the computer, its peripherals, stationery etc. Software resources would include the programs running on these computers and the liveware would include the human beings required to operate the system and make it functional.

Thus these resources make an important component of any system. For instance, a Banking system cannot function without the required stationery like cheque books, pass books etc. such systems also need computers to maintain their data and trained staff to operate these computers and cater to the customer requirements.

System Components cont...d

ii. Procedures

Every system functions under a set of rules that govern the system to accomplish the defined goal of the system. This set of rules defines the procedures for the system. For instance, the Banking systems have their predefined rules for providing interest at different rates for different types of accounts.

System Components cont...d

iii. Data/Information

Every system has some predefined goal. For achieving the goal the system requires certain inputs, which are converted into the required output. The main objective of the System is to produce some useful output. Output is the outcome of processing. Output can be of any nature e.g. goods, services or information.

However, the Output must conform to the customer's expectations. Inputs are the elements that enter the system and produce Output. Input can be of various kinds, like material, information, etc.

System Components cont...d

iv. Intermediate Data

Various processes process system's Inputs. Before it is transformed into Output, it goes through many intermediary transformations. Therefore, it is very important to identify the Intermediate Data. For example, in a college when students register for a new semester, the initial form submitted by student goes through many departments. Each department adds their validity checks on it.

Finally the form gets transformed and the student gets a slip that states whether the student has been registered for the requested subjects or not. It helps in building the System in a better way. Intermediate forms of data occur when there is a lot of processing on the input data. So, intermediate data should be handled as carefully as other data since the output depends upon it.

System Components cont...d

v. Processes

The systems have some processes that make use of the resources to achieve the set goal under the defined procedures. These processes are the operational element of the system.

For instance in a Banking System there are several processes that are carried out. Consider for example the processing of a cheque as a process. A cheque passes through several stages before it actually gets processed and converted. These are some of the processes of the Banking system. All these components together make a complete functional system.

Characteristics of a system

Systems also exhibit certain features and characteristics, some of which are:

i. Objective

This is a predefined goal of a system. A system cannot exist without a defined objective. For example an organization would have an objective of earning maximum possible revenues, for which each department and each individual has to work in coordination.

ii. Standards

It is the acceptable level of performance for any system. Systems should be designed to meet standards. Standards can be business specific or organization specific.

For example take a sorting problem. There are various sorting algorithms. But each has its own complexity. So such algorithm should be used that gives most optimum efficiency. So there should be a standard or rule to use a particular algorithm. It should be seen whether that algorithm is implemented in the system.

Characteristics of a system cont...d

iii. Environment

It is the surrounding of a particular system. Every system whether it is natural or man made co-exists with an environment. It is very important for a system to adapt itself to its environment. Also, for a system to exist it should change according to the changing environment.

For example, we humans live in a particular environment. As we move to other places, there are changes in the surroundings but our body gradually adapts to the new environment. If it were not the case, then it would have been very difficult for human to survive for so many thousand years.

iv. Feed Back

Feedback is an important element of systems. The output of a system needs to be observed and feedback from the output taken so as to improve the system and make it achieve the laid standards.

Characteristics of a system cont...d

v. Boundaries and Interfaces

Every system has defined boundaries within which it operates. Beyond these limits the system has to interact with the other systems.

For instance, Personnel system in an organization has its work

domain with defined procedures. If the financial details of an employee are required, the system has to interact with the Accounting system to get the required details.

Interfaces are another important element through which the system interacts with the outside world.

System interacts with other systems through its interfaces. Users of the systems also interact with it through interfaces. Therefore, these should be customized to the user needs. These should be as user friendly as possible.

Classifications of System

i. Physical vs Abstract System

Physical systems are tangible entities that we can touch and feel. These may be static or dynamic in nature. For example, take a computer center. Desks and chairs are the static parts, which assist in the working of the center. Static parts don't change. The dynamic systems are constantly changing.

Computer systems are dynamic system. Programs, data, and applications can change according to the user's needs.

Abstract systems are non-physical entities or conceptual that may be formulas, representation or model of a real system.

ii. Open vs Closed System

Open systems interact with their environment to achieve their targets. Things that are not part of the system are environmental elements for the system. Depending upon the interaction with the environment, systems can be divided into two categories, open and closed.

Classifications of System cont...d

Open systems: Systems that interact with their environment. Practically most of the systems are open systems. An open system has many interfaces with its environment. It can also adapt to changing environmental conditions. It can receive inputs from, and delivers output to the outside of system. An information system is an example of this category.

Closed systems: Systems that don't interact with their environment. Closed systems exist in concept only.

iii. Man made Information System

The main purpose of information systems is to manage data for a particular organization. Maintaining files, producing information and reports are few functions. An information system produces customized information depending upon the needs of the organization. These are usually **formal, informal, and computer based**.

Formal Information Systems: It deals with the flow of information from top management to lower management. Information flows in the form of memos, instructions, etc. But feedback can be given from lower authorities to top management.

Informal Information Systems: Informal systems are employee based. These are made to solve the day to day work related problems.

Computer-Based Information Systems: This class of systems depends on the use of computer for managing business applications.

Information system

Information System: A collection of interrelated components that collect, process, store, and disseminate information needed to complete business tasks. Three major information systems are:

1. **Transaction processing system:** TPS processes business transaction of the organization. Transaction can be any activity of the organization. Transactions differ from organization to organization. For example, take a railway reservation system. Booking, canceling, etc. are all transactions. Any query made to it is a transaction.

However, there are some transactions, which are common to almost all organizations. Like employee new employee, maintaining their leave status, maintaining employees accounts, etc.

2. **Management information system:** These systems assist middle management in problem solving and making decisions. They use the results of transaction processing and some other information also. It is a set of information processing functions. It should handle queries as quickly as they arrive. An important element of MIS system is database.
3. **Decision support system:**

These systems assist higher management to make long term decisions. These type of systems handle unstructured or semi structured decisions.

A decision is considered unstructured if there are no clear procedures for making the decision and if not all the factors to be considered in the decision can be readily identified in advance.

Data & Information

- **Data**: Streams of raw facts representing events such as business transactions.
- **Information**: Clusters of facts that is meaningful and useful to human beings in the processes such as making decisions.
- **Another definition:** Information is the product of data processing. Information is interrelated data. Information is equivalent to finished goods produced after processing the raw material. The information has a value in decision making.

System Analysis and Design

Systems development can generally be thought of as having two major components: Systems analysis and Systems design.

System analysis, is the process of investigating a system, identifying problems, and using the information to recommend improvements to the system.

System design is the process of planning a new business system or one to replace or complement an existing system.

Analysis specifies what the system should do. ***Design*** states how to accomplish the objective.

System Analysis and Design

- In business, System Analysis and Design refers to the process of examining a business situation with the intent of improving it through better procedures and methods.
- System analysis and design relates to shaping organizations, improving performance and achieving objectives for profitability and growth. The emphasis is on systems in action, the relationships among subsystems and their contribution to meeting a common goal.

System analyst

The system analyst is the person (or persons) who guides through the development of an information system .

Role of System Analyst differs from organization to organization. Most common responsibilities of System Analyst are following

1) System analysis

It includes system's study in order to get facts about business activity. It is about getting information and determining requirements. Here the responsibility includes only requirement determination, not the design of the system.

2) System analysis and design:

Here apart from the analysis work, Analyst is also responsible for the designing of the new system/application.

3) Systems analysis, design, and programming:

Here Analyst is also required to perform as a programmer, where he actually writes the code to implement the design of the proposed application.

Due to the various responsibilities that a system analyst requires to handle, he has to be multifaceted person with varied ***skills*** required at various stages of the life cycle. In addition to the technical knowhow of the information system development a system analyst should also have the following knowledge.

- **Business knowledge**: As the analyst might have to develop any kind of a business system, he should be familiar with the general functioning of all kind of businesses.
- **Interpersonal skills**: Such skills are required at various stages of development process for interacting with the users and extracting the requirements out of them
- **Problem solving skills**: A system analyst should have enough problem solving skills for defining the alternate solutions to the system and also for the problems occurring at the various stages of the development process

Users of the system

The **system end users** of the system refer to the people who use computers to perform their jobs, like desktop operators.

End users can be divided into various categories:

- **Hands-on users**: They actually interact with the system. They are the people who feed in the input data and get output data.

For instance; Like person at the booking counter of a gas authority. This person actually sees the records and registers requests from various customers for gas cylinders.

- **Indirect end users :** They do not interact with the systems hardware and software. However, these users benefit from the results of these systems. These types of users can be managers of organization using that system.
- **Users with management responsibilities for application systems:** They oversee investment in the development or use of the system.
- **Senior managers:** They are responsible for evaluating organization's exposure to risk from the systems failure.

Review questions

1. Define the term ‘System’.
2. What are the various elements of system?
3. Identify two systems in your surroundings.
4. What is system analysis and design?
5. What are the roles of system analyst?
6. Make a list of traits that a system analyst should have.
7. Will the responsibility of a system analyst vary according to:

(a)Organization size(for example small or large business)?

(b) Type of organization(business, government agency, non-profit organization)?

8. Differentiate between

a)Open and closed system

b) Physical and abstract

9. Main aim of an information system is to process _____.

10. Transaction processing, _____ , and decision support system are three types of information system.

11. State true or false

a) Decision support system is for middle level management.

b) Closed systems don't interact with their environment.

c) Transaction processing system handles day-to-day operations of the organization.

d) Management information system deals with strategic information of organization.

e) Problem solving and interpersonal skills are desirable for system analyst